

Public value management in digital transformation: a scoping review

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Abstract

Purpose – The aim of this study was to map and scope a body of literature on the interplay between public value management (PVM) and digital transformation (DT), clarify the concept of PVM in DT, and identify knowledge gaps by using a scoping review methodology.

Design/methodology/approach – A scoping review was conducted, drawing on the framework proposed by Arksey and O'Malley (2005) and Levac *et al.* (2010). A total of 54 publications on the interplay between PVM and DT were selected for the final analysis from the Web of Science, Scopus, and Dimensions databases.

Findings – We found that PVM in DT is a relatively unexplored topic and that additional research is needed to focus on the role of emerging technologies such as artificial intelligence (AI) and blockchain, especially in the context of local and regional governments. We also found that researchers have a limited conceptual understanding of PVM and DT.

Originality/value – This article makes two main contributions. First, although PVM and DT have contributed to public value research, our findings show a need for more systematic knowledge of these complementary but distinct approaches. Second, this study provides a valuable addition to the developing body of research using the scoping review methodology in PAM literature.

Keywords Public value management, Public value theory, Digital transformation, Scoping review, Public managers, Digital technology

Paper type Literature review

1. Introduction

Public sector reforms have resulted in paradigmatic changes and managerial implications for public administration and management (PAM) over the past few decades (Alford *et al.*, 2017). In the search for a new way of thinking in the public sector, the Weberian model of traditional public administration was challenged in the 1980s by New Public Management (NPM; Hood, 1991). Increased complexity problems, fragmentation of the public sector and an emphasis on efficiency and effectiveness led to strong criticism of NPM (Christensen, 2012; Reiter and Klenk, 2019). As a result, NPM was ultimately declared “dead” (Dunleavy *et al.*, 2006), and several post-NPM concepts emerged. Kim (2020) and Alford and Hughes (2008) argue that there is no commonly accepted term for the recent post-NPM discussion, and scholars have introduced various approaches to public administration, such as new public service



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(NPS; Denhardt and Denhardt, 2000), public value management (PVM; Moore, 1995; Stoker, 2006), digital-era governance (DEG; Dunleavy *et al.*, 2006) and new public governance (NPG; Osborne, 2006; for more details, see also Christensen, 2012; Kim, 2020; Torfing *et al.*, 2020). These new approaches are built on diverse theoretical and epistemological foundations. According to Greve (2015, p. 63), “DEG and digitalization are about legitimacy through transparency, PVM is about value creation, performance and results by innovative public managers, and NPG and involvement are about organizing resources through networks, collaboration, partnerships, and engaged citizens.” However, there is still some overlap among these new ideas and agendas, such as an emphasis on public value and how to deal effectively with networks of stakeholders in policymaking or public service delivery (Alford and Hughes, 2008; Bryson *et al.*, 2014; Torfing *et al.*, 2020).

Public value theory, as outlined by Moore (1995), has been especially successful in providing an alternative to the NPM narrative (Panagiotopoulos *et al.*, 2019), and Stoker’s (2006) PVM model has been acknowledged as a new paradigm and successor to NPM (Alford and Hughes, 2008; O’Flynn, 2007; Rutgers, 2015). Moreover, the role of public managers in PVM has been redefined from implementers of political plans to creators of collective value who deploy resources (money and authority) to fulfil citizens’ desires (Greve, 2015).

In recent years, there has been growing interest in the relationship between PVM and DT. Scholars have argued that the PVM approach provides an overarching framework for post-competitive, collaborative networks that accounts for the complex transformations fostered by digital technologies in public sector management (Panagiotopoulos *et al.*, 2019), such as digital transformation (DT), which is the focus of this article. Others have argued that digital technology is central to contemporary and succeeding waves of public sector reform (Dunleavy *et al.*, 2006; Greve, 2015). For example, van Veenstra and Janssen (2012) claim that information technology (IT) use is crucial for PVM to create public value for citizens, increase transparency and legitimisation, and strengthen the government’s role through collaboration in networks to achieve their goals. A recent empirical study showed that high-quality digital government services in a developing country increased flexibility and simplified processes for citizens (Buyannemekh *et al.*, 2024), and that a user-centred approach to value creation can influence the adoption of e-services by citizens (Lopes *et al.*, 2019).

Previous research has demonstrated that different technologies, such as the internet and artificial intelligence (AI), have improved how governments are organised and communicate with external stakeholders to support and enhance government operations and service delivery (Haug *et al.*, 2023). Although this process is often referred to as DT, many scholars consider it an ambiguous term (Mergel *et al.*, 2019). However, it can generally be understood as a constant change in the public sector due to its broad scope and complexity, and technology is a primary driver of transformative efforts (Haug *et al.*, 2023). The Organization for Economic Co-operation and Development (OECD, 2014) describes DT as a transition from analogue government through e-government – in which information and communication technology (ICT) is used to facilitate processes – to digital government.

Although PVM and DT refer to public sector reform in the digital age that addresses sector and community challenges (Stoker, 2006; Lips, 2020) and shifts the focus from public service production to public service creation (e.g. Criado and Gil-Garcia, 2019; Haug *et al.*, 2023; Lindquist, 2022; Panagiotopoulos *et al.*, 2019), it is still unclear how these two approaches relate to each other. There is also a lack of systematic empirical research on PVM (Bryson *et al.*, 2014; Torfing *et al.*, 2020) or how digital-era tools and social technologies further the value-creation process in the public sector (Lindquist, 2022; Hartley *et al.*, 2017). Therefore, the aim of this study was to map and scope a body of literature on the interplay between PVM and DT, clarify the concept of PVM in DT, and identify knowledge gaps. The research questions that guided this study were as follows:

RQ1. How has the PVM paradigm evolved in DT?

RQ2. How are the concepts of PVM and DT understood in the literature?

RQ3. What are the basic, motor, niche, emerging and declining research themes related to PVM in DT?

To answer these questions, we conducted a scoping review ([Arksey and O'Malley, 2005](#); [George et al., 2023](#); [Levac et al., 2010](#)) of 54 studies. To show the conceptual structure of PVM in DT, we built a thematic map using Bibliometrix ([Aria and Cuccurullo, 2017](#)) to identify basic, motor, niche, emerging, and declining themes. The findings of this study contribute to the public value literature by clarifying the development and relationships between PVM and DT, especially how digital technology has changed the behaviour of public managers in public value creation and suggesting avenues for future research. We also believe that this article provides a useful contribution to the evolving scoping review methodology in the PAM literature.

This article is organised as follows: The next section presents the methodological framework of the scoping review. We then present the findings related to the three research questions presented above. The final section discusses the study's main results and suggests avenues for future research.

2. Methodology

The interdisciplinarity and complexity of the concepts of PVM and DT and their interplay create the need to clarify existing knowledge on the development of PVM in DT. A scoping review methodology was used to identify and map the breadth of data available on PVM in DT, clarify the concept of PVM in DT and examine the research trends related to this topic ([Arksey and O'Malley, 2005](#)). According to [Munn et al. \(2018, p. 2\)](#), a scoping review is “an ideal tool to determine the scope or coverage of a body of literature on a given topic and give a clear indication of the volume of literature and studies available as well as an overview (broad or detailed) of its focus”. Therefore, a scoping review has a broader scope and is more objective and reliable than a traditional literature review ([George et al., 2023](#)). As [Levac et al. \(2010\)](#) and [Arksey and O'Malley \(2005\)](#) recommend, our scoping review was conducted according to the following procedure (see also [Figure 1](#)):

- (1) Identifying the research question: The first step in the search strategy was to identify literature reviews on PVM and DT in the PAM literature. Despite there being few studies identifying public value or value creation in a digital context, no scoping or systematic literature reviews have been conducted on this topic. Our searches were conducted in July and August 2023 using Web of Science, Scopus and the Dimensions database. We sought literature containing the combinations of the keywords digit*, e—*, elect*, tech*, internet*, smart*, G2C, G2G, G2B and G2E with “public value*” (see Supplementary material [appendix](#) for full overview). In the keywords, we used a broad selection strategy to obtain a comprehensive overview of the DT literature. Thus, we did not limit the search to specific technologies – e.g. AI, the Internet of Things (IoT), big data or social media – and practical solutions (e.g. platforms or living labs). Instead, we used terms such as “digital” and “e-government” because they involve applications of different technologies in the public sector.
- (2) Identifying relevant studies: This stage aimed to identify, as exhaustively as possible, a set of studies that would enable us to answer our research questions. In the first step, we identified 3,005 studies. There was no restriction on the publication period. After removing duplicates using Deduplick, we were left with 1,779 studies from three databases. In the next step, we excluded 1,337 studies by screening the titles,

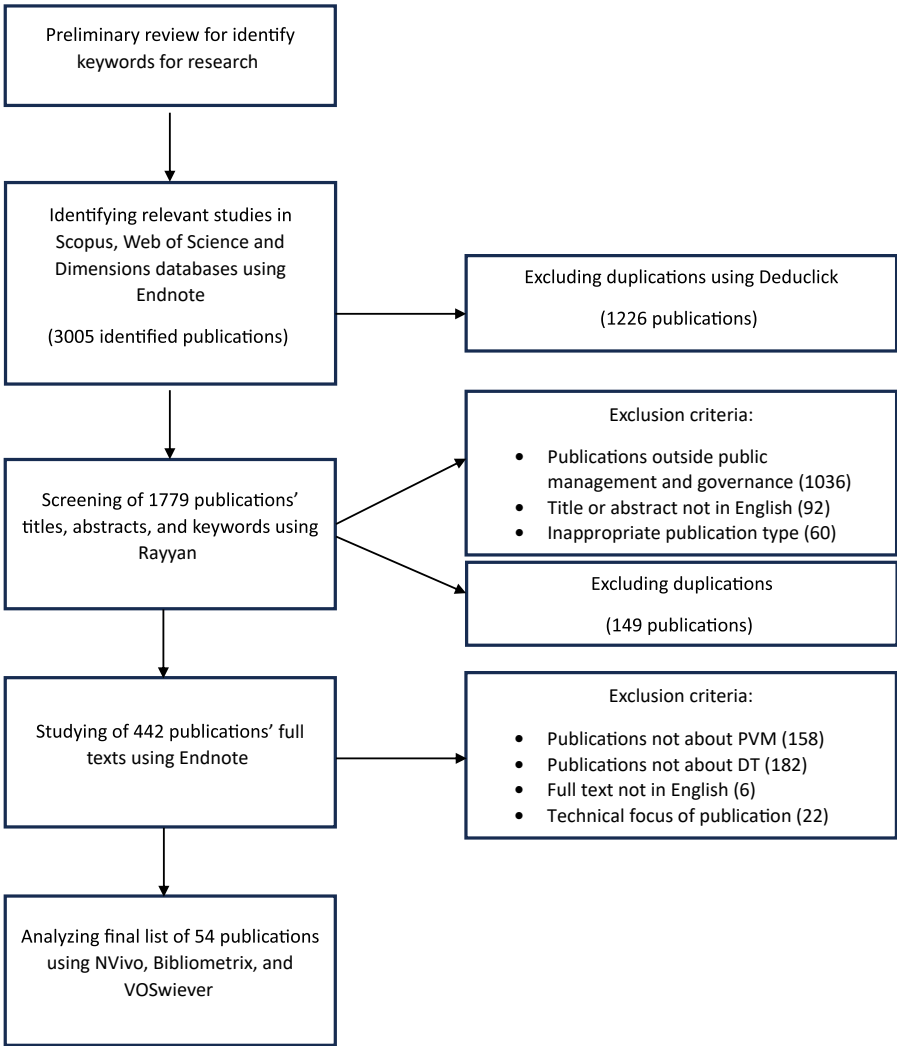


Figure 1.
Flow chart for the
scoping review process

Source(s): Authors' own work

abstracts, and keywords (see the flow chart in [Figure 1](#) for details) according to the adopted exclusion criteria ([Table 1](#)). We then manually identified 149 other duplicates by checking titles, abstracts and keywords that Deduclink had not removed. Out of the 422 studies, 368 were excluded, and 54 were included for full-text reading and analysis (see [Figure 1](#)). Three researchers carried out this three-step process.

- (3) Study selection was based on two screening steps: Between August and September 2023 were the titles, abstracts and keywords screened using Rayyan (<https://www.rayyan.ai/>) ([Ouzzani et al., 2016](#)). We excluded studies with abstracts in languages other than English. We included studies that focused on public value in a digital context. Discrepancies during screening were resolved by consensus among the

Table 1.
Inclusion and
exclusion criteria

Criterion	Inclusion	Exclusion
Time period	To increase coverage, no starting date was used. Research up to September 2023 was included	No restrictions
Language	English	Non-English studies
Type of publication	Peer-reviewed journal articles, conference proceedings, book chapters	Journal and book reviews
Phenomenon of interest	The interplay between PVM and DT	Publications that only mentioned PVM in the text; Publications where DT is only mentioned in the text
Field of study	Social sciences	Outside social sciences
Type of study	Both theoretical and empirical	No restrictions
Source(s): Authors' own work		

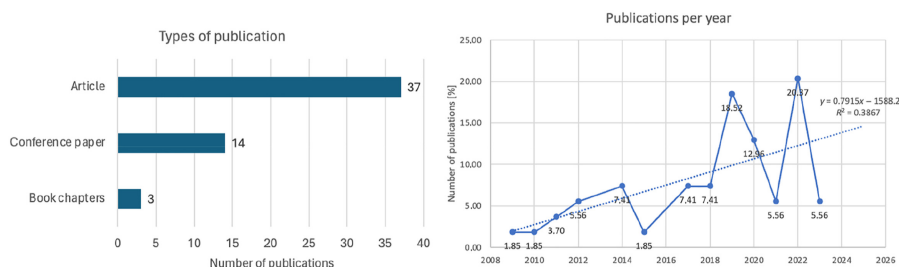
researchers. Second, using Endnote for reference management, we screened full texts to verify whether the selected studies allowed us to obtain answers to the research questions. For this reason, we used several inclusion and exclusion criteria (see Table 1). In this stage, we eliminated studies that did not address the interplay between PVM and DT – i.e. those that mentioned PVM or DT but were not directly concerned with them. Studies on DT outside the social sciences were also excluded.

- (4) Charting the data: The 54 selected studies were coded using NVivo to find answers to our research questions. We used the following codes: definitions of PVM and DT, theoretical perspectives, study topics, public sector dimensions, digital technology types, study types (empirical or theoretical/conceptual), methods (qualitative, qualitative or mixed methods), context level (individual, organisational or inter-organisational), country and level of government (local, national or international). The bibliometric analysis was visualised using Excel. To extract key information from the analysed studies, we used Bibliometrix and VOSviewer to prepare charts. Based on the co-occurrence of the authors' keywords, we identified the first thematic clusters within the research area of PVM in DT. The thematic map then helped us concretise the identified networks and, in particular, to compare them in a matrix approach to obtain a detailed co-occurrence analysis. This procedure enabled the assessment of research topics in four clusters: basic, niche, motor and emerging/declining.

3. Results

3.1 Bibliometric analysis

The results presented in Figure 2 show studies published in the last 15 years that were included in the analysis. Most of the identified 54 studies were peer-reviewed articles ($N = 37$),

**Figure 2.**
Number and type of
identified publications

and one-third of them were published in interdisciplinary journals, such as *Government Information Quarterly* ($N = 10$) and the *International Journal of Information Management* ($N = 2$). Fourteen studies were published as conference papers. The selected book chapters were in the minority ($N = 3$). Three years after [Stoker \(2006\)](#) developed the PVM model, the first two conference papers on PVM in the context of DT were published in 2009 and 2010; the first peer-reviewed article was published in 2011. There was a slight increase in the number of published studies in the first 10 years and a decrease in 2015 and 2021. The decline in 2015 may have occurred due to the emergence of other post-NPM paradigms in the public management literature focusing on value, such as public service logic (PSL; [Osborne et al., 2015](#); [Osborne et al., 2013](#)). The decline in 2021 can be explained by increased scholarly interest in issues related to the COVID-19 pandemic. We should also note that our results only concern the interplay between PVM and DT, not both concepts independently.

Most of the included studies were empirical (64.8%); the remaining 35.2% were theoretical or conceptual. In the empirical studies, research was carried out in the field of various government functions listed in the Classification of the Functions of Government (COFOG), and they considered several digital technologies (see [Figure 3](#)).

Slightly more than half of the studies selected for analysis (51.84%) defined government functions, focusing on five of the 10 government functions presented in the COFOG ([Eurostat, 2019](#)). In most cases, general public services dominated, while the remaining four functions were investigated sporadically. Then, the analysed technologies were dominated by a general approach to ICT (24.07%) with AI (14.81%). Digital platforms and the IoT were mentioned less frequently, and only in one case was research conducted on blockchain technology.

The bibliometric analysis shows that the identified studies on PVM in DT create a suitable research background for conducting a substantive analysis of this topic. Below, we present the analysis in light of our research questions.

3.2 Development of PVM in DT

This section presents the evolution of the PVM paradigm in DT. Bibliometrix is an R-based application that allows for combining multiple data describing a given research area. It was used to illustrate which references were most often cited in the included studies, their authors and what research topics in the field of PVM in DT have developed most ([Figure 4](#)).

The results presented in [Figure 4](#) show that [Moore's \(1995\)](#) seminal book *Creating Public Value* introduced the concept of PVM. However, the studies by [Kelly et al. \(2002\)](#) and [Kearns \(2004\)](#) also made a significant contribution to the literature on PVM in DT, as they described the premises and principles of public value creation in the context of digital-based public sector reforms. The studies of [O'Flynn \(2007\)](#), [Cordella and Bonina \(2012\)](#) and [Bannister and Connolly \(2014\)](#) further contributed to the development of the PVM and DT approaches. O'Flynn's article mainly discusses the evolution and importance of PVM and contrasts it with

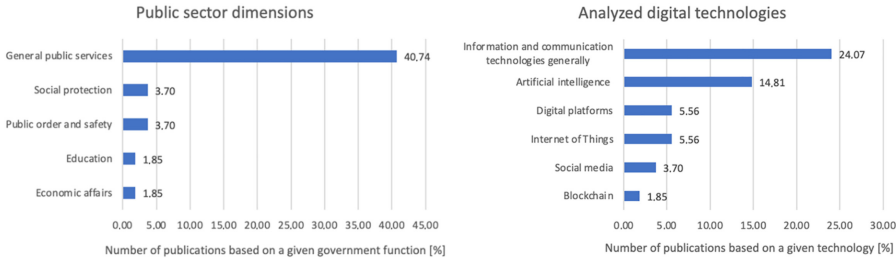
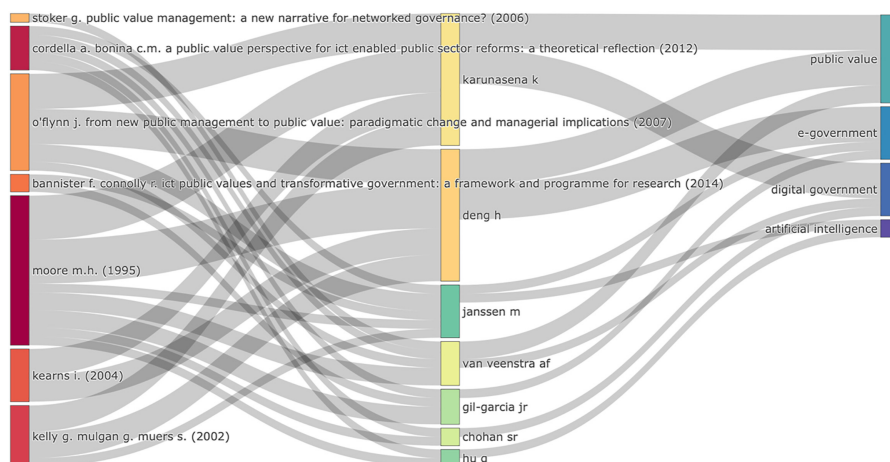


Figure 3.
Public sector
dimensions and digital
technologies

Source(s): Authors' own work



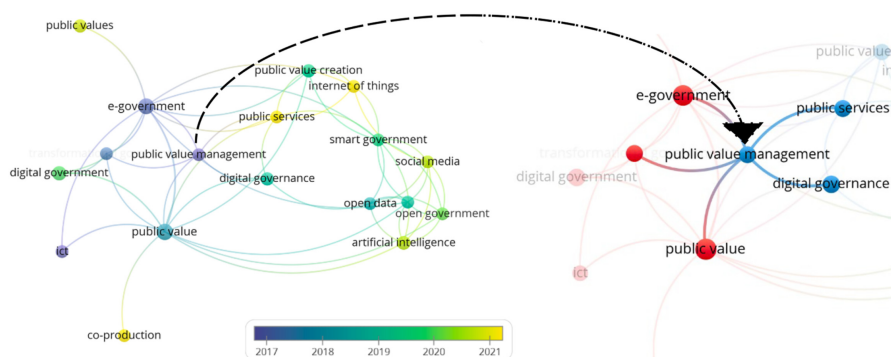
Source(s): Authors' own work using Bibliometrix (Aria and Cuccurullo, 2017)

Figure 4.
Development of PVM
and DT research

NPM. The other two studies (Cordella and Bonina, 2012; Bannister and Connolly, 2014) began the development of research on PVM in DT by showing the importance of ICT in public value research. However, our analysis shows that Karunasena and Deng (2009, 2010, 2011, 2012) made the most significant contribution to the development of the interplay between PVM in DT (Figure 4). These authors directly address and connect Stoker's PVM model and DT approach to their research.

Our results also show that PVM has evolved through the two main stages of DT (see OECD, 2014): e-government and digital government. In particular, emerging technologies such as AI and social media have been used in research to illustrate the development of the PVM in DT. We used VOSviewer to illustrate the relationships between these two approaches (see Figure 5).

The position of the PVM paradigm in Figure 5 highlights its interplay with DT (and its related concepts). PVM is linked with all the main stages of DT development (e-government, digital government and transformational government), digital governance focusing on the management and operation of digital services, and digital technology types (IoT, social media and AI).



Source(s): Authors' own work using VOSviewer 1.6.20

Figure 5.
Overlay visualization
of PVM in the context
of DT

Our results show that research combining PVM with e-government and ICT before 2017 has provided the basis for the development of PVM in DT (Figure 5). The most popular issues concerning PVM in DT between 2017 and 2020 were public value, digital governance, digital government, open data, public value creation and smart governments. In recent years, the focus has shifted to social media and modern technologies such as AI and IoT as well as issues related to stakeholder involvement, including co-production and co-creation of public services.

3.3 Understanding PVM and DT

Although all the included studies focused on PVM, only a few included a definition or explanation of PVM (see Table 2). These definitions portray PVM as one of the emerging public management paradigms focusing on public managers’ key role in exploring and exploiting public value in co-operation with other stakeholders – such as citizens, community organisations, policymakers, businesses, politicians and non-governmental organisations – to achieve favourable outcomes. This co-operation should result in determining what constitutes public value for a given community and how it can be created in a DT context (Lindgren and van Veenstra, 2018; Pang *et al.*, 2014; Pereira *et al.*, 2020; van Veenstra and Janssen, 2012; van Veenstra and Timan, 2022).

Although several studies lacked a definition of PVM, “public value” was defined in many. The focus on public value, reminding us that the government exists to serve society, is seen as a real achievement of the PVM paradigm (Torfing *et al.*, 2020). Scholars describe this concept primarily as citizens’ collective expectations of public services and government, which should be met through legal regulations and the actions of politicians (e.g. Twizeyimana and Andersson, 2019; Valle-Cruz, 2019). Chatfield and Reddick (2020) emphasise that PVM constitutes a shift from a focus on effectiveness and efficiency, which characterises the NPM paradigm, to citizens’ engagement in public processes based on collaborative governance.

Table 2.
PVM definitions

Public value management	Source
... “is concerned with the dynamic assessment of what constitutes public value and how this can be created in networks of stakeholders”	van Veenstra and Timan (2022, p. 190)
... “argues public managers’ key role is determination and pursuit of public values through engagement and deliberation with elected politicians, stakeholders, and citizens”	Madan and Ashok (2023, p. 2)
... “a strategic approach for the creation of value. In this context, managers are considered strategic actors that work in building necessary capabilities and stakeholder networks that enable them to create value”	Pereira <i>et al.</i> (2020, p. 605)
... “an attempt of creating an integrated and holistic vision on value for society by collaboration of public and private parties in networks”	van Veenstra and Timan (2022) and Van Veenstra and Janssen (2012, p. 190)
... “relies on a stakeholder conception of legitimacy in its governance arrangements. The fundamental idea is that for a decision to be legitimate or for a judgement to be made, it is necessary to have all the stakeholders involved”	Pang <i>et al.</i> (2014, p. 194)
... “engagement and delivery with networked collaborations including non-profits, community organizations, and other actors. ... ‘public value’ can be achieved in idiosyncratic ways, which may vary by service, levels of government, and community”	Lindquist (2022, p. 552)

The authors of the included studies often referred to the three components of public value (Kelly *et al.*, 2002): (1) services; (2) outcomes; and (3) trust, legitimacy and confidence. According to Wang *et al.* (2021), this classification is helpful, as it refers to both internal processes (between managers and employees) and social impact and interactions between public managers and service stakeholders. This classification creates room for the implementation of digital technologies to facilitate relationships between these three components using digital platforms (van Noordt and Misuraca, 2020). For example, Ndlovu *et al.* (2023) argue that public value arises from the benefits of DT that support interactions between the government and citizens. However, finding a common definition that covers PVM and DT, or even PVM and digital technology, is difficult. This may be due to the fact that both PVM and DT are still evolving approaches.

DT encompasses and goes beyond the transition from e-government (digitalisation) to digital government (DG), which radically changes service delivery practices through the use of digital technology (see Table 3). DT is also considered an effective way to meet the expectations of the stakeholders of public organisations (Janowski, 2015; Scupola and Mergel, 2022; Tangi *et al.*, 2020).

Digital transformation	Source
... "is a holistic effort to revise core processes and services of government beyond the traditional digitization efforts. It evolves along a continuum of transition from analogue to digital to a full stack review of policies, current processes, and user needs and results in a complete revision of the existing and the creation of new digital services. The outcome of digital transformation efforts focuses among others on the satisfaction of user needs, new forms of service delivery, and the expansion of the user base"	Scupola and Mergel (2022, p. 4)
... "relates to a new way of delivering public services through the integration of new and innovative technologies and changing needs that require a user-driven administration and clear value delivery"	Pereira <i>et al.</i> (2020, p. 604)
... "efforts aim to move beyond the e-government efforts of creating better service delivery for citizens and businesses and realize public sector reform"	Van Veenstra and Janssen (2012, p. 187)
... "is an organizational change strongly mediated by digital technologies, making the government more agile, flexible, integrated, cooperative, efficient, transparent, and centred in the real needs of society"	Coutinho and Freitas (2021, p. 5)
... "is the latest wave of technological change and not as a coherent reform movement: it is furthering diverse and often competing values, leading to new balances simultaneously across government"	Lindquist (2022, p. 549)

Table 3.
DT definitions

The above definitions show that DT leads to a shift in public service provision methods, resulting in a change in the scope and type of public value. These changes are permanent and radical and create new approaches to PVM (Tangi *et al.*, 2020). Bannister and Connolly (2014) and van Noordt and Misuraca (2020) suggest that digital technology has a particular impact on public value in terms of internal organisational processes, public service provision processes and value creation for society. However, other scholars emphasise that access to technology itself is not a determinant of DT; what is crucial here is how public managers and other public servants use these technologies in managing their activities and creating public value (e.g. Pereira *et al.*, 2020; Scupola and Mergel, 2022; Tangi *et al.*, 2020).

3.4 Research themes of PVM and DT

Bibliometrix's thematic map tool was applied to check the validity of the research questions regarding the interplay between PVM and DT (Aria and Cuccurullo, 2017). We used centrality and density rank to measure the relevance degree of the research themes (X-axis)

and their development degree (Y-axis). [Figure 6](#) presents the thematic map, which classifies PVM and DT research into four thematic categories ([Zamani *et al.*, 2022](#)):

- (1) Basic: transversal and well-understood research themes
- (2) Motor: key, well-developed and relevant research themes
- (3) Niche: themes with low centrality and high density, meaning they are less important in research on PVM in DT than the two previous categories
- (4) Emerging/declining: weakly developed and marginal themes.

The circles in [Figure 6](#) indicate the research clusters. Each theme is characterized by two measures: centrality (the number of relational ties a research constituent has in a network) and density (the internal strength of the network). The clusters consist of different topics; for instance, the blue cluster consists of four motor themes considered together in the literature: public administration, the public sector, e-governance and digital governance.

The results in [Figure 6](#) indicate that the smart city is an important and well-developed research theme for PVM in DT. This topic is widely discussed in the literature from the perspective of city transformation ([Pereira *et al.*, 2020](#)) and the role of citizens in public value co-creation ([Cao and Kang, 2022](#)).

At the intersection of the basic and motor themes, there is also a large cluster relating to public values, e-government, government data processing and public services. This cluster emphasises the importance of open government data to generate public value for society by governments together with other stakeholders, such as non-governmental actors ([Chu and Tseng, 2018](#); [Coutinho and Freitas, 2021](#)). Studies on this theme also emphasise that technology is an important mechanism for boosting public value in the delivery process ([Twizeyimana and Andersson, 2019](#); [Valle-Cruz, 2019](#); [Zhang and Kimathi, 2022](#)).

In the cluster with the highest density and centrality, digital governance and e-governance issues are discussed in the context of public administration. Studies in this cluster emphasise the important role of digital technologies, such as ICT, in public administration and the conscious implementation of social and organisational shifts towards using these technologies effectively ([Chu *et al.*, 2017](#); [Misuraca and Viscusi, 2015](#); [van Veenstra and Timan, 2022](#)).

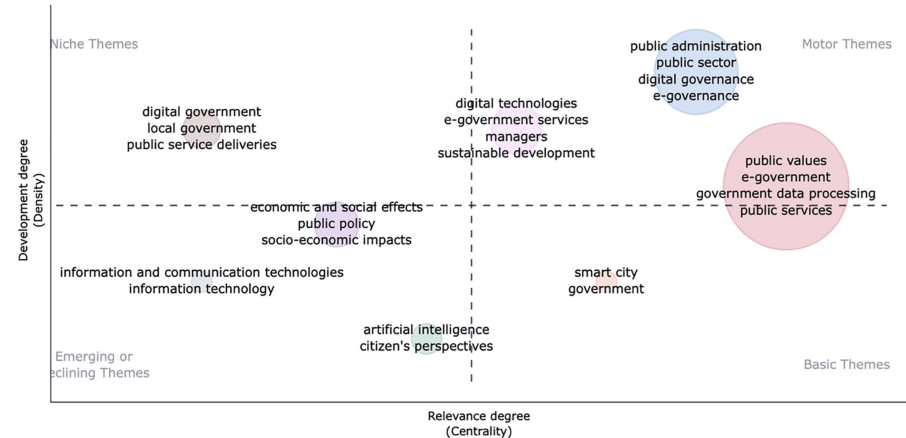


Figure 6.
Thematic map

Source(s): Authors' own work using Bibliometrix ([Aria and Cuccurullo, 2017](#))

The critical role of public managers involved in PVM in DT can also be found among the motor themes. These studies often refer to how public managers use digital technologies and e-government in pursuit of sustainable development (Janowski *et al.*, 2018; Karunasena and Deng, 2011; Pereira *et al.*, 2020).

Among the niche themes, only one cluster deals with public service delivery at the local level using the digital government approach. This research is scattered and needs more scholarly attention. In this theme, the researchers examine the impact of digital technologies on public value co-creation (Casula *et al.*, 2022) and attempt to develop an understanding of the causal paths leading from implementing digital administration to improving public service provision. Moreover, scholars use the public value perspective to research co-operation between local governments wherein they attempt to adapt technological changes and implement modern public services (Puron-Cid, 2017; Soe and Drechsler, 2018).

Among the emerging and declining themes, we identified three clusters: two emerging (the socio-economic impacts of ICT capabilities on public value and the use of AI in research on PVM in DT from citizens' perspectives), and one declining (the use of ICT and IT in PVM in DT research). These are new issues that arise from the general themes of public value creation in digital governance. Researchers argue that DT is expected to have an economic impact and address social challenges, but this research has not been extensively explored (Misuraca and Viscusi, 2015; Samoilenko and Osei-Bryson, 2019; van Veenstra and Timan, 2022). Similarly, studies emphasising the role of AI in PVM need to be explored empirically (Luna *et al.*, 2019; Wang *et al.*, 2021).

4. Discussion and conclusion

This article identified and analysed the main sources and evidence on PVM in DT. Through the three research questions, we (1) demonstrated how the PVM paradigm evolved in DT; (2) described how the concepts of PVM and DT are understood in the literature; and (3) identified the basic, motor, niche and emerging/declining research themes related to PVM in DT. We also conducted a bibliometric analysis of the 54 studies selected for the scoping review.

Our findings demonstrate that research exploring the interplay between PVM and DT developed slowly over the first 10 years since 2009 and that it is still a relatively new and immature topic in the PAM literature. Early PVM and DT studies were mainly conceptual and lacked empirical evidence (e.g. Bryson *et al.*, 2014). However, recent research has demonstrated the usefulness of PVM in DT, which is mainly dominated by single-case studies and focuses on general public services supported by ICT and AI technologies. Therefore, we argue that a comparative perspective combining qualitative and quantitative methods and focusing on different emerging technologies would strengthen the research stream that focuses on the interplay between PVM and DT.

Although PVM has been acknowledged as a new post-NPM paradigm (Alford and Hughes, 2008; O'Flynn, 2007; Rutgers, 2015), other competing and complementary paradigms have emerged amid the post-NPM debate (Alford *et al.*, 2017; Osborne *et al.*, 2013; Torfing *et al.*, 2020). Based on our findings, we assume that the emergence of PSL, a new paradigm of value creation in the public management literature, may have contributed to the decline in the number of studies published in 2015 and 2021 (Figure 1). The latest review of PSL shows a research expansion of mainly conceptual articles on PSL (since 2015), followed by empirical research (since 2021; Zyzak and Martinussen, 2024). However, the roles of public managers in the PVM and PSL models differ. In PVM, public managers play a dominant role in creating public value, while PSL boosts the role of service users in value creation. Future studies on the interplay between these paradigms are therefore recommended; in particular, researchers contributing to the post-NPM debate should carefully consider the role of managers and other stakeholders in value creation in DT, as this would help public managers apply these models in practice.

While research on PVM started with [Moore's \(1995\)](#) seminal book, researchers in other disciplines (e.g. IT, computer science, management and political science) adopted the PVM model developed by [Stoker \(2006\)](#) to study how digital technologies can facilitate public value creation ([van Veenstra and Janssen, 2012](#)). Nevertheless, our analysis revealed a lack of scientific rigour in defining PVM and DT and the interplay between them, which may lead to confusion in applying the PVM model in practice. Thus, this article contributes to filling this research gap by offering a definition of PVM in DT: the key role of a public manager using PVM in DT is to endorse and enable the use of modern digital technologies (e.g. AI, blockchain and the IoT) to create public value in collaboration with various stakeholders, including citizens.

Our analysis reveals that digital technologies are tools/means for public managers that enable them to create public value together with diverse stakeholders. However, due to the growing role of emerging technologies such as AI, blockchain and IoT, we found that ICT is no longer a sufficient technology for public value creation ([Chohan and Hu, 2020](#); [Chohan et al., 2023](#); [Criado and Gil-Garcia, 2019](#); [Fatima et al., 2022](#)). Research has shown that ICT is gradually being replaced or supplemented by new technologies that offer different applications and usefulness for PVM. Given this imperative, there remains a need for empirical research to understand what types of technologies are necessary for public managers to enable public value creation in specific dimensions of the public sector, such as healthcare, education, security, etc. Moreover, digital public infrastructure should be modernized and prepared for the use of modern technologies in creating public value. This requires an integrated rather than fragmented approach to digital policymaking.

Finally, this article identified the mainstream and niche topics in the debate on PVM in DT that can lead to entirely new avenues of research. Our thematic map in [Figure 6](#) shows that studies on smart cities have given rise to research on the PVM paradigm in DT ([Pereira et al., 2020](#)). The smart city perspective ([Gil-Garcia et al., 2016](#); [Meijer and Bolívar, 2016](#)) enables public managers to open up to innovation and flexibility based on smart technologies to create common solutions with diverse stakeholders. Moreover, topics that have received sufficient attention and provided empirical studies on PVM in DT focus on the role of managers in delivering sustainable e-services ([Janowski et al., 2018](#)) and creating public value in governments based on data processing ([Chu and Tseng, 2018](#)).

One important finding that caught our attention is the decline in interest in the use of ICT in PVM in DT research. This can be explained by the increase in research on emerging new technologies such as AI, blockchain and IoT. Although the empirical research is still weak, this technological wave encourages transformative practices in public sector organizations. Moreover, this shift may have resulted from the ongoing research indicating a transition from e-government to digital government ([Ndlovu et al., 2023](#); [OECD, 2014](#); [Puron-Cid, 2017](#); [van Veenstra and Timan, 2022](#)). This change has implications for practitioners, as several structural and cultural barriers persist in public sector organizations that inhibit the use of new technologies. Therefore, a better dialogue between theory and practice is needed. Finally, a niche topic – i.e. poorly explored and narrow but potentially yielding interesting results in terms of PVM development – is the interconnection between digital government and local government and the provision of public services ([Soe and Drechsler, 2018](#)).

Our study, like other scoping reviews, has some limitations because the purpose is to provide breadth rather than depth of information on a particular topic. First, we limited the included studies to those published in English; therefore, our results are generalisable to literature reviews in English. Moreover, this study is constrained in part by methodological choices, such as the selection of keywords and the screening, coding, and analysis techniques we used. However, three scholars have been involved in all the data collection and analysis phases to mitigate these limitations. Overall, we believe that this scoping review provides a unique overview of research on PVM in DT.

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Appendix

Searched keyword	Concept reference samples	Sample publications
Digit*	Digitization	Dunleavy <i>et al.</i> (2006) , Greve (2015) , Lindgren <i>et al.</i> (2019) , Lips (2020) , Mergel <i>et al.</i> (2019)
	Digitalization	
	Digital transformation	
	Digital-Era Governance	
	Digital public services	
	Digital governance	
	Digital government	
	Digital democracy	
	Digital inclusion/exclusion	
	Digital collectivism	
	Digital divide	
	Digital privacy	
	Digital literacy	
	Digital technologies	
	Digital threats	
	Digital security	
	Digital society	
	Digital citizenship	
	Digital age	
E—*	e-gov	Lips (2020) , Sandoval-Almazan and Gil-Garcia (2012) , Karunasena <i>et al.</i> (2011)
	e-governance	
	e-government	
	e-democracy	
	e-procurement	
	e-service	
	e-citizens	
	e-participation	
	e-consultation	
Electr*	e-policymaking	Lips (2020) , Milakovich (2021) , Misuraca and Viscusi (2015)
	Electronic government	
	Electronic governance	
	Electronic communication	
	Electronic democracy	
	Electronic community	
Tech*	Electronic document	Charalabidis <i>et al.</i> (2022) , El-Haddadeh <i>et al.</i> (2019) , Milakovich (2021) , Velsberg <i>et al.</i> (2020)
	Technology	
	Information and communication technologies (ICT)	
	Digital technologies	
	Technological change	
	Technological dominance	
	Technological ethic	
	Technology perspective	
Internet*	Internet governance	De Gregorio and Radu (2022) , Lips (2020) , Milakovich (2021)
	Internet connectivity	
	Internet use	
	Internet readiness	
	Internet service providing	
	Internet revolution	

Table A1.
Searching keywords
combination with
“public value”

(continued)

Table A1.

Searched keyword	Concept reference samples	Sample publications
Smart*	Smart government Smarter governing Smart technologies Smart citizens Smart state Smart contracts Smart approaches	Charalabidis <i>et al.</i> (2022) , Lips (2020) , Meijer and Bolívar (2016) , Milakovich (2021) , Noveck (2015)
G2*	G2C (Government to citizen) G2G (Government to government) G2B (Government to business) G2E (Government to employee)	Charalabidis <i>et al.</i> (2022) , Lips (2020) , Milakovich (2021)
Source(s): Authors' own work		

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